

Tarun Vinay Gujjar

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Dear Hiring Team,

I am writing to express my strong interest in the Applications Programmer Associate position within the Robotics Department at Michigan Engineering. As a recent Master's graduate in Computer Science from George Mason University, where I achieved a GPA of 3.83, I am inspired by Michigan's mission of developing technology that serves humanity, an ideal that closely aligns with my own motivation for creating efficient, human-centered software systems.

In my recent role as a Software Engineer, I designed and optimized RESTful APIs using Python, improved system efficiency, and automated CI/CD pipelines with AWS and Jenkins to enable reliable and faster deployments. I also collaborated with cross-functional teams to troubleshoot integration issues and enhance user experience, which directly aligns with the support and application enhancement goals outlined in this role. Earlier in my career, I developed data-driven tools in Python and Java to improve vehicle telematics workflows, implemented automated testing frameworks, and mentored new engineers, experiences that honed my technical problem-solving and collaboration skills. My proficiency in Python, JavaScript, SQL, and web application development, combined with hands-on experience in AWS, Docker, and Kubernetes, equips me to build scalable and secure applications while ensuring compliance and operational integrity.

I am particularly excited about contributing to Michigan's Robotics initiatives, where my background in automation, data systems, and software engineering can support innovative research and the department's educational objectives. I view this position as an opportunity to apply my technical expertise within an academic environment that values impact and collaboration. I am confident that my combination of technical skills, analytical thinking, and commitment to continuous improvement will strengthen the department's digital infrastructure and contribute to Michigan Engineering's vision of a people-first future.

Thank you for considering my application. I would be delighted to discuss how my skills and experiences align with your team's objectives.

Sincerely,

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EDUCATION

Master of Science in Computer Science

George Mason University, Fairfax, VA

Courses: Machine Learning, Deep Learning, Component-Based Software Development, Analysis of Algorithms

May 2025

GPA: 3.83/4.0

Bachelor of Engineering in Computer Science & Engineering

The National Institute of Engineering, Mysore, India

Courses: Object-Oriented Programming, Data Structures and Algorithms, Cloud Computing, Operating System

August 2021

GPA: 9.0/10.0

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, SQL, HTML, CSS, JavaScript.
- **Data Science & Analytics:** NumPy, Pandas, Matplotlib, Scikit-Learn, Data Visualization (Tableau, Power BI).
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS, DynamoDB), Docker, Kubernetes, Jenkins, CI/CD.
- **Version Control System & Collaboration:** Git, GitHub, GitLab, Bitbucket.
- **Testing & Automation:** Unit Testing, Integration Testing, Python Selenium, PyTest, API Testing (Postman).

WORK EXPERIENCE

Software Engineer Intern - Data Semantics, Fairfax, United States of America

July 2025 - Present

- Designed and developed RESTful APIs in Python with CRUD operations and authentication/authorization, improving integration efficiency by 50% and supporting scalable backend web services.
- Optimized API performance by refactoring queries and introducing caching strategies, reducing response time by 20% and improving user experience across modules.
- Collaborated with front-end and QA teams to document endpoints (Postman), create test cases, and ensure seamless integration, which reduced defect leakage during system testing.
- Implemented automated CI/CD pipelines on AWS (EC2, Lambda) with Jenkins and PyTest-based test suites, increasing release reliability and accelerating deployments by 30%.

Software Engineer - Daimler Truck Innovation Center India (DTICI), Bangalore, India

August 2021 - July 2023

- Developed and deployed Python and Java scripts using object-oriented design principles in a production environment to enhance data processing, resulting in a 50% increase in application usage across automobile modules.
- Implemented an automated testing framework using Python Selenium and unit tests, boosting efficiency by 40% and improving resource management with version control, bug tracking, and documentation.
- Diagnosed and resolved critical application issues in a Linux environment through code review and log analysis, and drove Agile improvements that elevated customer satisfaction in 2 weeks.
- Led onboarding sessions for 10 new software engineering hires, simplifying complex project workflows and emphasizing data quality to facilitate efficient knowledge transfer and collaboration.

PROJECTS

Cloud-Native Deployment System:

- Created a containerized web application using Docker, ensuring 100% portability across different platforms.
- Orchestrated deployment using Kubernetes, improving scalability by 60% and reducing downtime.
- Built a CI/CD pipeline using Jenkins, integrating AWS Cloud Infrastructure, and optimizing for scalability, accelerating deployment time by 40% and ensuring seamless builds and testing.
- Integrated GitHub, enhancing collaborative development, version tracking, and reducing code conflicts by 30%.

Student Performance Analysis Model:

- Pre-processed 10,000+ student records by handling missing values, normalizing data, and refining feature selection.
- Designed and deployed predictive models using ensemble learning, regression modeling techniques, linear programming, and nonlinear optimization, achieving an R^2 score of 0.936 for accurate performance evaluation.
- Fine-tuned regression techniques, improving model efficiency by 30% and enhancing prediction reliability.

Placement Cell Application:

- Engineered a Kafka-based queuing system to handle high-volume concurrent job applications, boosting request throughput and ensuring fault-tolerant, real-time processing.
- Optimized database performance by implementing advanced indexing and query tuning, enabling fast retrieval of thousands of student records and cutting query latency by approximately 40%.